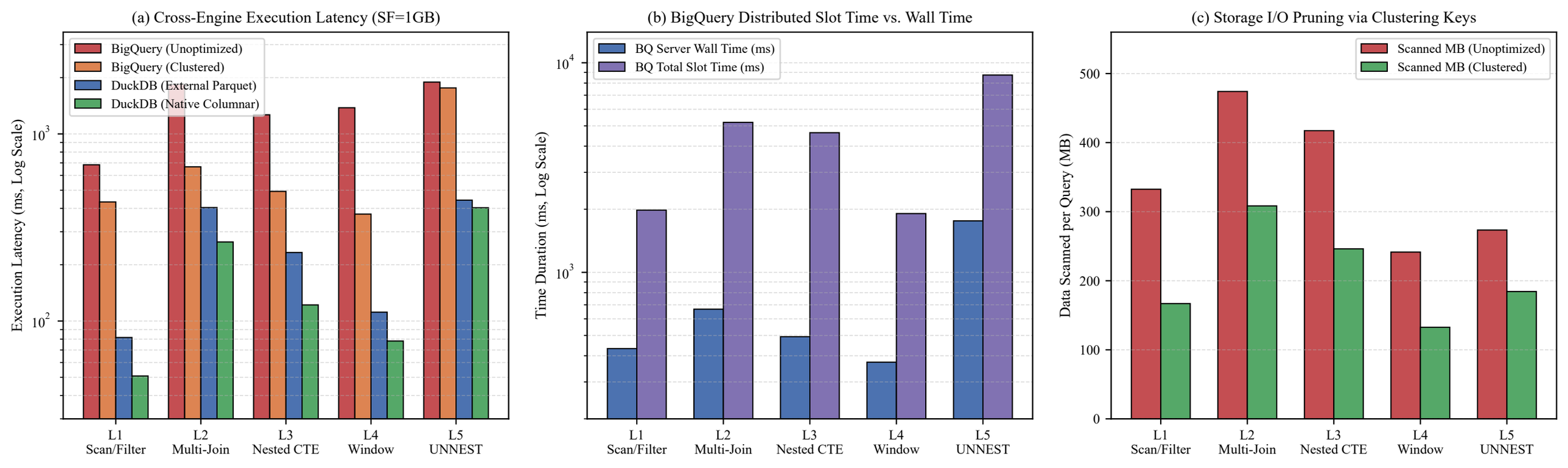
(a) Cross-Engine Execution Latency (SF=1GB)
BigQuery (Unoptimized)
BigQuery (Clustered)
DuckDB (External Parquet)
DuckDB (Native Columnar)
Execution Latency (ms, Log Scale)
10^3
10^2
L1 Scan/Filter
L2 Multi-Join
L3 Nested CTE
L4 Window
L5 UNNEST
(b) BigQuery Distributed Slot Time vs. Wall Time
BQ Server Wall Time (ms)
BQ Total Slot Time (ms)
Time Duration (ms, Log Scale)
10^4
10^3
L1 Scan/Filter
L2 Multi-Join
L3 Nested CTE
L4 Window
L5 UNNEST
(c) Storage I/O Pruning via Clustering Keys
Scanned MB (Unoptimized)
Scanned MB (Clustered)
Data Scanned per Query (MB)
0
100
200
300
400
500
L1 Scan/Filter
L2 Multi-Join
L3 Nested CTE
L4 Window
L5 UNNEST